

Homework 7

Deadline: April 14, 2022

E 6.4 b

6.4 (b) The vapour pressure of a substance at 20.0°C is 58.0 kPa and its enthalpy of vaporization is 32.7 kJ mol^{-1} . Estimate the temperature at which its vapour pressure is 66.0 kPa .

E 6.5 a

6.5 (a) The molar volume of a certain solid is $161.0\text{ cm}^3\text{ mol}^{-1}$ at 1.00 atm and 350.75 K , its melting temperature. The molar volume of the liquid at this temperature and pressure is $163.3\text{ cm}^3\text{ mol}^{-1}$. At 100 atm the melting temperature changes to 351.26 K . Calculate the enthalpy and entropy of fusion of the solid.

E 6.8 a

6.8 (a) When benzene freezes at 5.5°C its density changes from 0.879 g cm^{-3} to 0.891 g cm^{-3} . Its enthalpy of fusion is 10.59 kJ mol^{-1} . Estimate the freezing point of benzene at 1000 atm .

E 6.11 b

6.11 (b) The boiling point of hexane is 69.0°C . Estimate (a) its enthalpy of vaporization and (b) its vapour pressure at 25°C and 60°C .